Multi-Corner Static Timing Analysis Report

vsdbabysoc - SKY130 HD (sky130_fd_sc_hd) - OpenROAD-flow-scripts

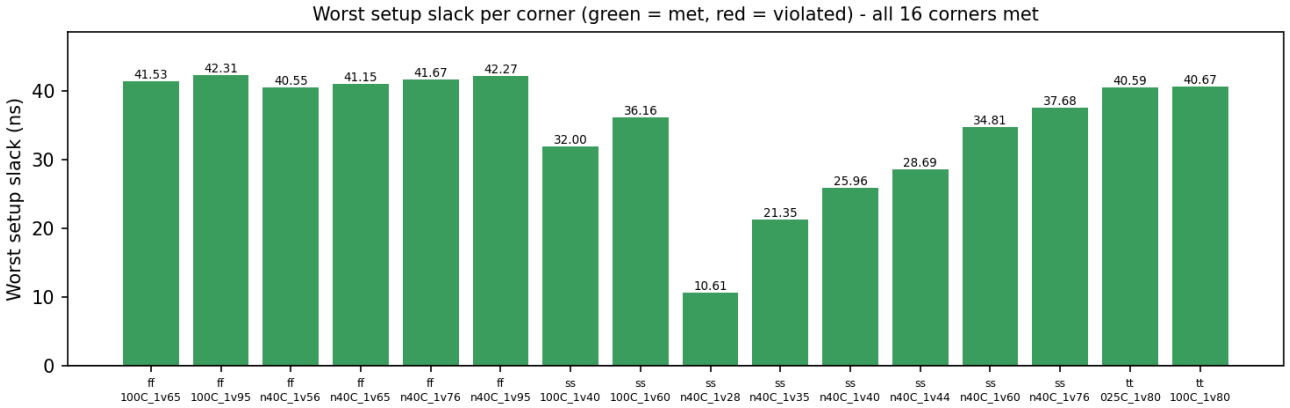
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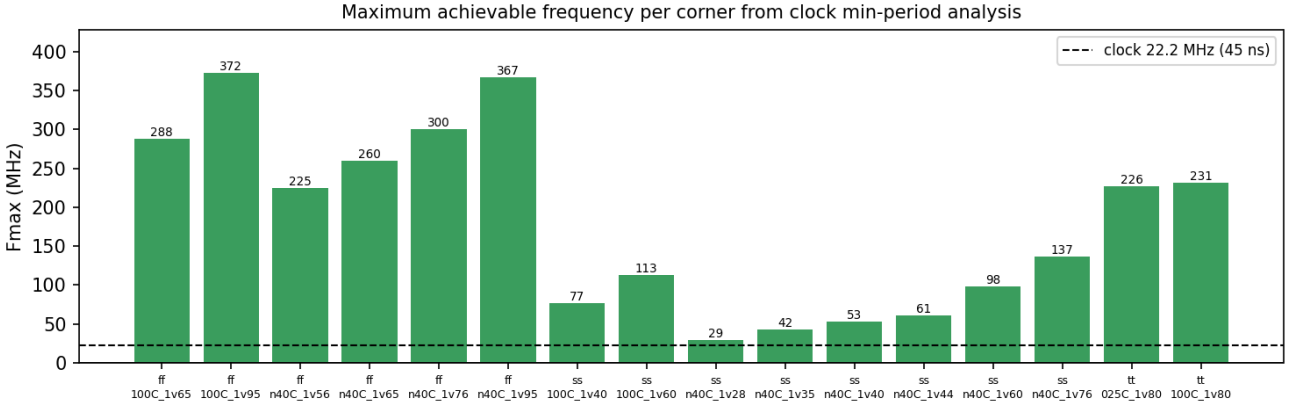
Report information: Setup and hold timing sign-off of the routed vsdbabysoc design across all 16 PVT corners (ff, tt, ss) using OpenSTA in OpenROAD, with a 45 ns clock and global-routing parasitics. The design was analysed after an in-place slew/capacitance repair (162 instances resized, 50 buffers inserted), and all 16 corners meet timing with zero rule violations.

Result Summary - All 16 Corners (45 ns clock, after repair)

#	Corner	Process / Temp / V	Setup slack	Hold slack	Setup WNS	Setup TNS	Skew	Min period	Fmax (MHz)	Rule viol.	Status
1	ff_100C_1v65	FF / +100 C / 1.65 V	41.53	0.27	0.00	0.0	0.15	3.47	288.3	0	MET
2	ff_100C_1v95	FF / +100 C / 1.95 V	42.31	0.21	0.00	0.0	0.12	2.69	372.4	0	MET
3	ff_n40C_1v56	FF / -40 C / 1.56 V	40.55	0.31	0.00	0.0	0.19	4.44	225.0	0	MET
4	ff_n40C_1v65	FF / -40 C / 1.65 V	41.15	0.27	0.00	0.0	0.16	3.85	259.5	0	MET
5	ff_n40C_1v76	FF / -40 C / 1.76 V	41.67	0.24	0.00	0.0	0.14	3.33	300.1	0	MET
6	ff_n40C_1v95	FF / -40 C / 1.95 V	42.27	0.20	0.00	0.0	0.11	2.73	366.6	0	MET
7	ss_100C_1v40	SS / +100 C / 1.40 V	32.00	0.93	0.00	0.0	0.47	13.00	76.9	0	MET
8	ss_100C_1v60	SS / +100 C / 1.60 V	36.16	0.66	0.00	0.0	0.33	8.85	113.0	0	MET
9	ss_n40C_1v28	SS / -40 C / 1.28 V	10.61	1.63	0.00	0.0	1.27	34.39	29.1	0	MET
10	ss_n40C_1v35	SS / -40 C / 1.35 V	21.35	1.31	0.00	0.0	0.86	23.65	42.3	0	MET
11	ss_n40C_1v40	SS / -40 C / 1.40 V	25.96	1.11	0.00	0.0	0.71	19.04	52.5	0	MET
12	ss_n40C_1v44	SS / -40 C / 1.44 V	28.69	1.01	0.00	0.0	0.59	16.31	61.3	0	MET
13	ss_n40C_1v60	SS / -40 C / 1.60 V	34.81	0.68	0.00	0.0	0.39	10.19	98.1	0	MET
14	ss_n40C_1v76	SS / -40 C / 1.76 V	37.68	0.51	0.00	0.0	0.29	7.32	136.6	0	MET
15	tt_025C_1v80	TT / +25 C / 1.80 V	40.59	0.33	0.00	0.0	0.18	4.41	226.5	0	MET
16	tt_100C_1v80	TT / +100 C / 1.80 V	40.67	0.33	0.00	0.0	0.18	4.33	231.1	0	MET

All times in ns. Setup / hold slack are the worst slack of the respective path group. Status is MET only when both setup and hold are non-negative. Rule viol. is the number of pins violating max slew, max capacitance or max fanout. Result: 16 of 16 corners MET, 0 rule violations (was 1,560 violating pins before repair).





Multi-Corner Static Timing Analysis Report

Design: vsdbabysoc | PDK: sky130_fd_sc_hd | Tool: OpenROAD (OpenSTA) | Flow: OpenROAD-flow-scripts

1. Objective

Validate setup and hold timing and the slew/capacitance rules of the routed vsdbabysoc design at all 16 Liberty PVT corners of sky130_fd_sc_hd (fast, typical and slow). The database analysed is **5_route_repaired.odb**, i.e. the routed design after the in-place repair, with a **45 ns** clock.

2. Commands used

Corner analysis (run once per corner, output saved to sta_corners_rep/):

```
cd ~/Desktop/OpenROAD-flow-scripts/flow
sed 's#5_route.odb#5_route_repaired.odb#' sta_corner_45.tcl > sta_corner_rep.tcl
mkdir -p sta_corners_rep
for c in $(ls sta_corners_out | sed 's/\.rpt$//'); do
  CORNER=$c CORNER_LIB=${L}$c.lib /opt/openroad/bin/openroad -no_splash -exit sta_corner_rep.tcl \
      > sta_corners_rep/$c.rpt 2>&1
done

# summary table
for f in sta_corners_rep/*.rpt; do echo "$(basename $f .rpt) viol=$(sed -n '/CHECK TYPES/,$p' $f \
  | grep -c VIOLATED) $(grep -h 'worst slack' $f | tr '\n' ' ')"; done
```

Each report contains the worst setup path (max), the worst hold path (min), WNS/TNS, clock skew, *clk period_min* and *fmax*, and a CHECK TYPES section that lists slew/cap/fanout violations (empty means none). All reports show **PARASITICS: global_routing**.

Reading the 'worst slack' lines: **worst slack max** is the worst **setup** slack, and **worst slack min** is the worst **hold** slack. Positive slack means the check is met.

3. Results for all 16 corners (clock = 45 ns, after repair)

Corner	Setup slack (ns)	Hold slack (ns)	WNS/TNS	Rule violations	Min period (ns)	fmax (MHz)	Setup margin vs fmax
ff_100C_1v65	41.532	0.265	0.00 / 0.00	0	3.47	288.3	13.0x
ff_100C_1v95	42.314	0.210	0.00 / 0.00	0	2.69	372.4	16.8x
ff_n40C_1v56	40.555	0.311	0.00 / 0.00	0	4.44	225.0	10.1x
ff_n40C_1v65	41.147	0.273	0.00 / 0.00	0	3.85	259.5	11.7x
ff_n40C_1v76	41.668	0.241	0.00 / 0.00	0	3.33	300.1	13.5x
ff_n40C_1v95	42.273	0.203	0.00 / 0.00	0	2.73	366.6	16.5x
ss_100C_1v40	32.003	0.933	0.00 / 0.00	0	13.00	76.9	3.5x
ss_100C_1v60	36.155	0.661	0.00 / 0.00	0	8.85	113.0	5.1x
ss_n40C_1v28	10.609	1.633	0.00 / 0.00	0	34.39	29.1	1.3x
ss_n40C_1v35	21.351	1.309	0.00 / 0.00	0	23.65	42.3	1.9x
ss_n40C_1v40	25.962	1.113	0.00 / 0.00	0	19.04	52.5	2.4x
ss_n40C_1v44	28.689	1.009	0.00 / 0.00	0	16.31	61.3	2.8x
ss_n40C_1v60	34.807	0.681	0.00 / 0.00	0	10.19	98.1	4.4x
ss_n40C_1v76	37.681	0.508	0.00 / 0.00	0	7.32	136.6	6.1x
tt_025C_1v80	40.585	0.328	0.00 / 0.00	0	4.41	226.5	10.2x
tt_100C_1v80	40.673	0.331	0.00 / 0.00	0	4.33	231.1	10.4x

Summary. All 16 corners pass setup and hold with zero violations. The tightest setup corner is **ss_n40C_1v28** (+10.609 ns); the tightest hold corner is **ff_n40C_1v95** (+0.203 ns). The slowest corner needs a minimum period of 34.39 ns (fmax 29.08 MHz), which is why a 45 ns clock (22.2 MHz) is required for setup to pass at every corner.

4. Graphs

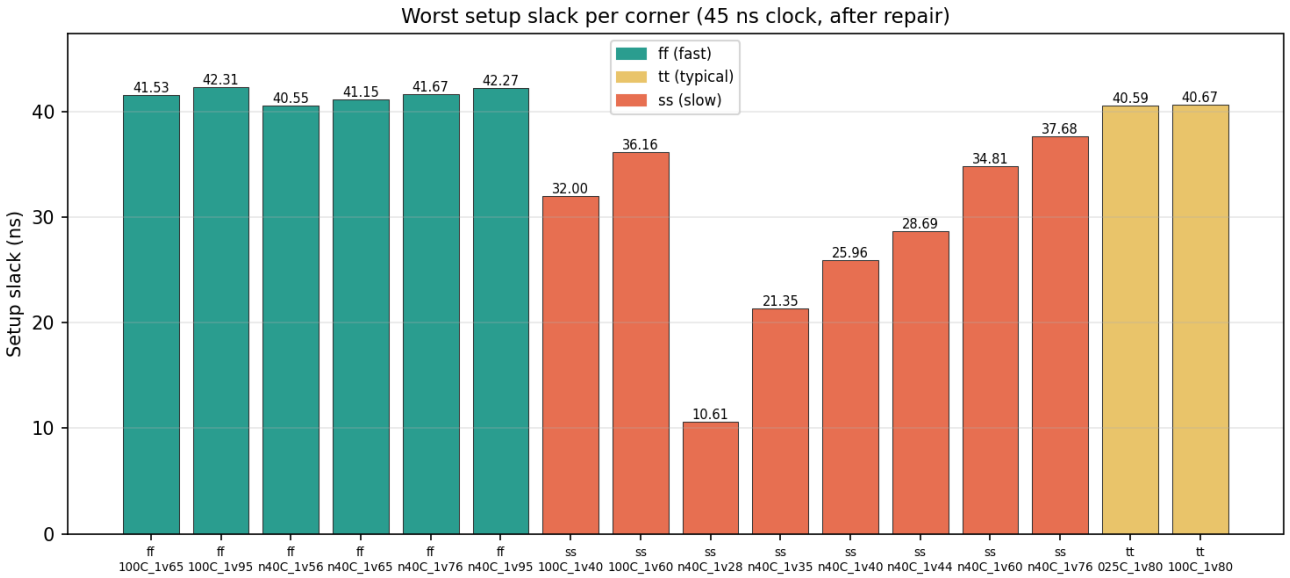


Figure 1 - Setup slack at every corner. The slow, low-voltage, cold corners are tightest.

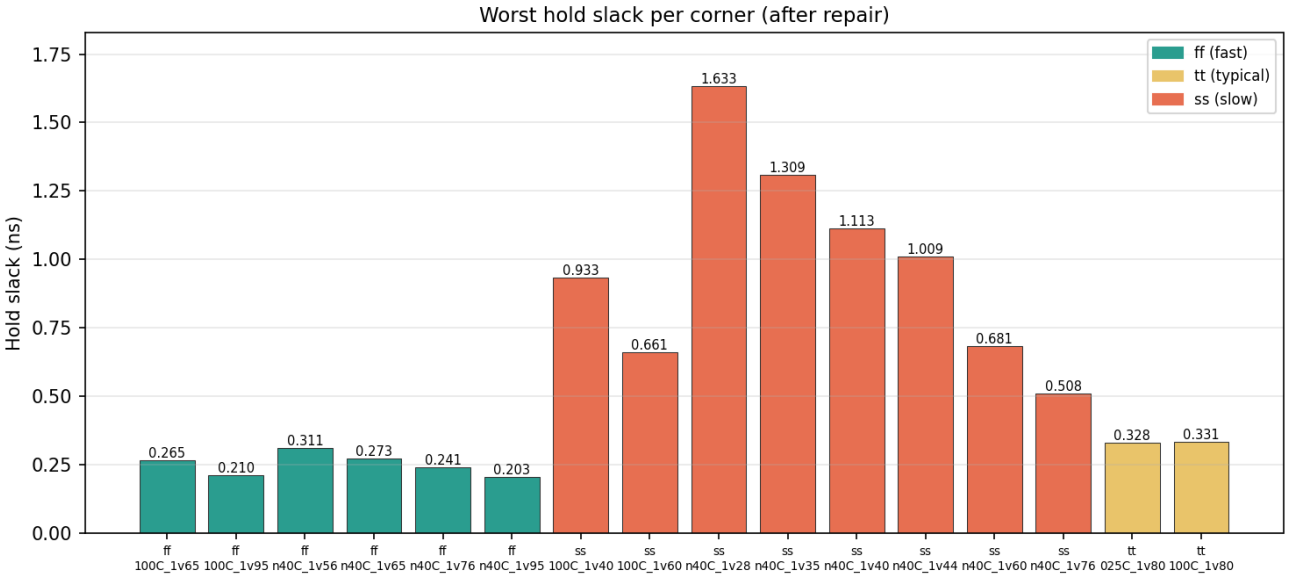


Figure 2 - Hold slack at every corner. Fast corners are tightest for hold; all are positive.

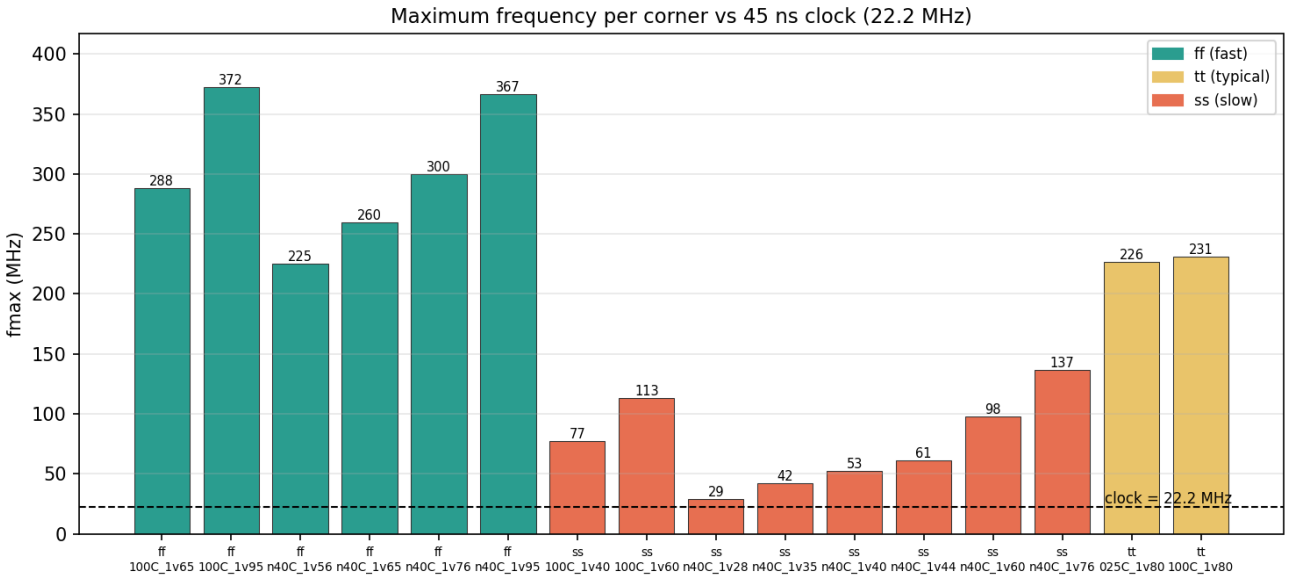


Figure 3 - fmax from each corner's report against the 22.2 MHz clock. Every corner is above the line.

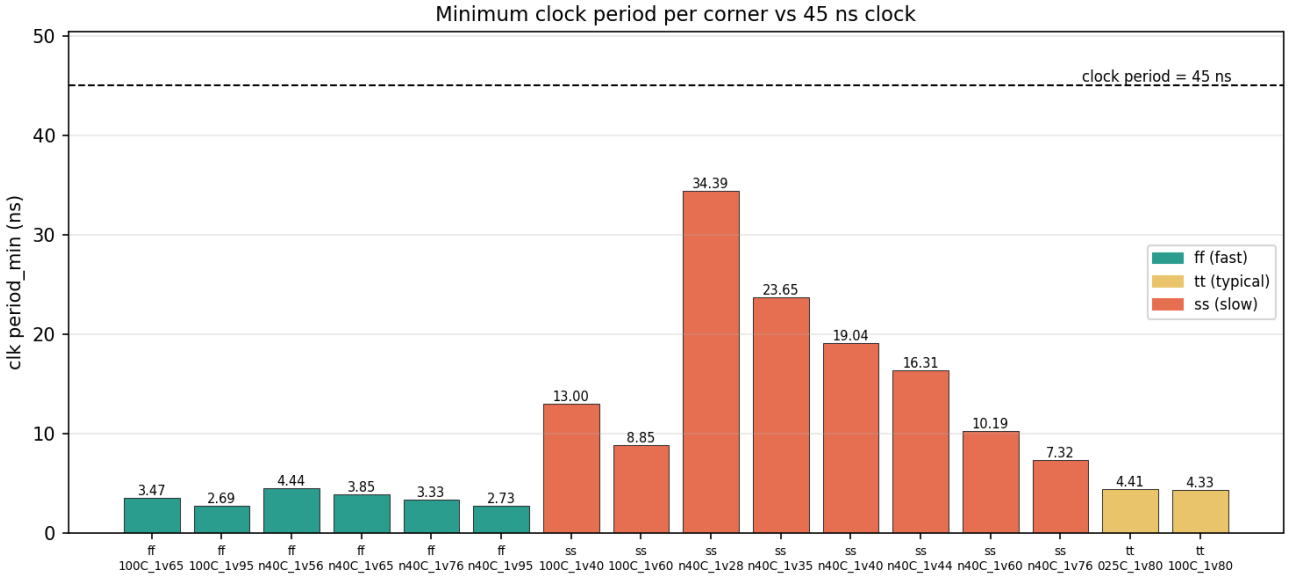


Figure 4 - Minimum clock period against the 45 ns period. ss_n40C_1v28 comes closest (34.39 ns).

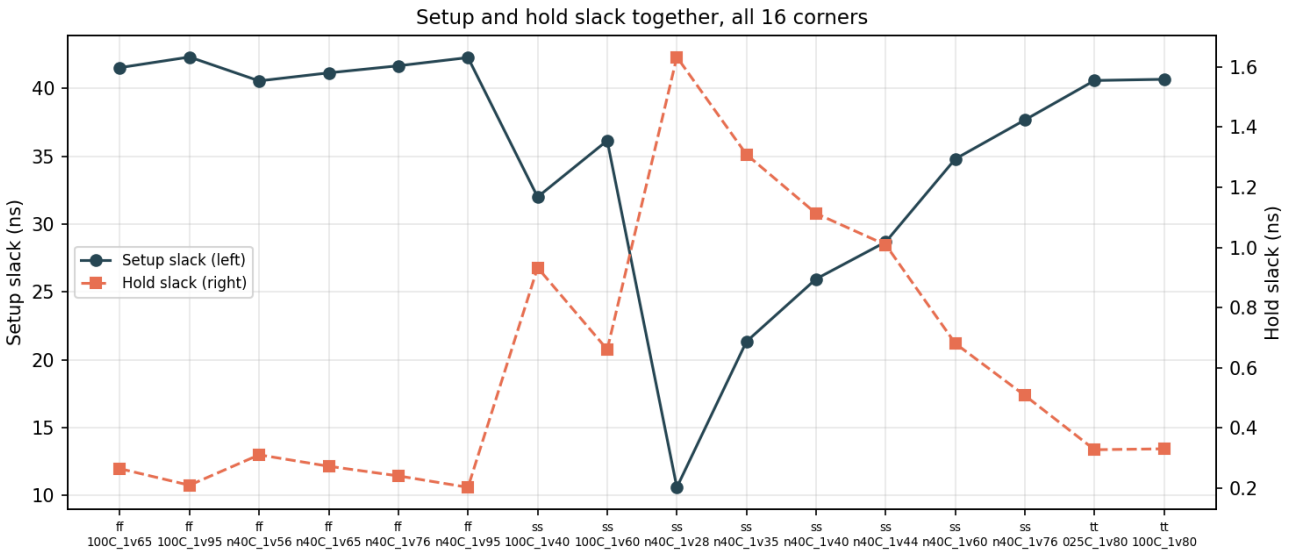


Figure 5 - Setup and hold slack together. Setup slack falls and hold slack rises as the corner gets slower.

Observations

1. **Setup is limited by the slow corners.** Setup slack ranges from +10.6 ns (ss_n40C_1v28) to +42.3 ns (ff_100C_1v95). Lower voltage and low temperature slow the cells (inverted temperature dependence at low voltage in sky130).

2. **Hold is limited by the fast corners.** Hold slack is smallest at ff_n40C_1v95 (+0.203 ns) and ff_100C_1v95 (+0.210 ns), and grows to +1.633 ns at ss_n40C_1v28.

3. **Speed spread.** fmax varies from 29.08 MHz to 372.36 MHz, a spread of about 12.8x between the worst and best corner.

5. Closing the slew/cap violations

Before the repair, the routed design had 1,560 slew/capacitance violating pins across 7 slow corners. The repair was run in place on the routed database and saved as a new file, **5_route_repaired.odb**, in two passes. The original routed database and the 11 ns reports were not modified.

Pass	Instances resized	Buffers inserted	Nets repaired
Repair A	155	33	74
Repair B	7	17	18
Total	**162**	**50**	**92**

Source: OpenROAD log lines RSZ-0039 (resized) and RSZ-0038 (buffers inserted).

Three-stage comparison

Metric	11 ns baseline	45 ns clock	45 ns + repair
Corners with setup violations	6	0	0
Corners with hold violations	0	0	0
Corners with slew/cap violations	7 (1,560 pins)	7 (1,560 pins)	0
Worst setup slack (ss_n40C_1v28)	negative	+1.22 ns	+10.61 ns
Hold slack (ss_n40C_1v28)	-	+1.83 ns	+1.63 ns

The 11 ns and 45 ns columns come from the earlier runs of this project, not from the files in this report. Only the 'after repair' column is backed by the per-corner reports that follow. 'Negative' for the 11 ns setup slack in ss_n40C_1v28 is inferred from the fact that its minimum period (34.39 ns) exceeds 11 ns.

6. Per-corner results

Each page below shows the key lines of the OpenSTA report for that corner: the worst setup path endpoints with arrival, required time and slack, the worst hold path, worst slack, WNS/TNS, clock skew and fmax. The complete path traces are in the saved report files (sta_corners_rep/*.rpt).

6.1 Corner ff_100C_1v65 (fast)

Setup slack **+41.532 ns** and hold slack **+0.265 ns**: both checks are met, with no rule violations. Minimum period 3.47 ns (fmax 288.3 MHz), so the 45 ns clock has 13.0x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_100C_1v65
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.653   data required time
                           4.121   data arrival time
-----------------------------------------------------------
                          41.532   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.054   data arrival time
                           0.789   data required time
-----------------------------------------------------------
                           0.265   slack (MET)

worst slack max 41.53
worst slack min 0.27
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.74 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.59 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.15 setup skew

clk period_min = 3.47 fmax = 288.34
=== CHECK TYPES (empty = none) ===
```

6.2 Corner ff_100C_1v95 (fast)

Setup slack **+42.314 ns** and hold slack **+0.210 ns**: both checks are met, with no rule violations. Minimum period 2.69 ns (fmax 372.4 MHz), so the 45 ns clock has 16.8x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_100C_1v95
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.554   data required time
                           3.240   data arrival time
-----------------------------------------------------------
                          42.314   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           0.859   data arrival time
                           0.650   data required time
-----------------------------------------------------------
                           0.210   slack (MET)

worst slack max 42.31
worst slack min 0.21
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.62 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.50 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.12 setup skew

clk period_min = 2.69 fmax = 372.36
=== CHECK TYPES (empty = none) ===
```

6.3 Corner ff_n40C_1v56 (fast)

Setup slack **+40.555 ns** and hold slack **+0.311 ns**: both checks are met, with no rule violations. Minimum period 4.44 ns (fmax 225.0 MHz), so the 45 ns clock has 10.1x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_n40C_1v56
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.796   data required time
                           5.241   data arrival time
-----------------------------------------------------------
                          40.555   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.297   data arrival time
                           0.986   data required time
-----------------------------------------------------------
                           0.311   slack (MET)

worst slack max 40.55
worst slack min 0.31
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   1.01 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.82 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.19 setup skew

clk period_min = 4.44 fmax = 224.99
=== CHECK TYPES (empty = none) ===
```

6.4 Corner ff_n40C_1v65 (fast)

Setup slack **+41.147 ns** and hold slack **+0.273 ns**: both checks are met, with no rule violations. Minimum period 3.85 ns (fmax 259.5 MHz), so the 45 ns clock has 11.7x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_n40C_1v65
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.716   data required time
                           4.570   data arrival time
-----------------------------------------------------------
                          41.147   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.149   data arrival time
                           0.875   data required time
-----------------------------------------------------------
                           0.273   slack (MET)

worst slack max 41.15
worst slack min 0.27
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.89 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.73 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.16 setup skew

clk period_min = 3.85 fmax = 259.52
=== CHECK TYPES (empty = none) ===
```

6.5 Corner ff_n40C_1v76 (fast)

Setup slack **+41.668 ns** and hold slack **+0.241 ns**: both checks are met, with no rule violations. Minimum period 3.33 ns (fmax 300.1 MHz), so the 45 ns clock has 13.5x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_n40C_1v76
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.645   data required time
                           3.978   data arrival time
-----------------------------------------------------------
                          41.668   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.018   data arrival time
                           0.776   data required time
-----------------------------------------------------------
                           0.241   slack (MET)

worst slack max 41.67
worst slack min 0.24
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.80 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.66 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.14 setup skew

clk period_min = 3.33 fmax = 300.08
=== CHECK TYPES (empty = none) ===
```

6.6 Corner ff_n40C_1v95 (fast)

Setup slack **+42.273 ns** and hold slack **+0.203 ns**: both checks are met, with no rule violations. Minimum period 2.73 ns (fmax 366.6 MHz), so the 45 ns clock has 16.5x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ff_n40C_1v95
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.561   data required time
                           3.288   data arrival time
-----------------------------------------------------------
                          42.273   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           0.860   data arrival time
                           0.657   data required time
-----------------------------------------------------------
                           0.203   slack (MET)

worst slack max 42.27
worst slack min 0.20
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.62 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.51 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.11 setup skew

clk period_min = 2.73 fmax = 366.64
=== CHECK TYPES (empty = none) ===
```

6.7 Corner ss_100C_1v40 (slow)

Setup slack **+32.003 ns** and hold slack **+0.933 ns**: both checks are met, with no rule violations. Minimum period 13.00 ns (fmax 76.9 MHz), so the 45 ns clock has 3.5x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_100C_1v40
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[3][29]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          46.344   data required time
                          14.341   data arrival time
-----------------------------------------------------------
                          32.003   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           2.980   data arrival time
                           2.047   data required time
-----------------------------------------------------------
                           0.933   slack (MET)

worst slack max 32.00
worst slack min 0.93
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   1.95 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.48 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.47 setup skew

clk period_min = 13.00 fmax = 76.94
=== CHECK TYPES (empty = none) ===
```

6.8 Corner ss_100C_1v60 (slow)

Setup slack **+36.155 ns** and hold slack **+0.661 ns**: both checks are met, with no rule violations. Minimum period 8.85 ns (fmax 113.0 MHz), so the 45 ns clock has 5.1x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_100C_1v60
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[3][29]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          46.054   data required time
                           9.900   data arrival time
-----------------------------------------------------------
                          36.155   slack (MET)

=== HOLD ===
Startpoint: core.CPU_reset_a2$_DFF_P_
Endpoint:   core.CPU_reset_a3$_DFF_P_
Path Group: clk    Path Type: min
                           2.157   data arrival time
                           1.495   data required time
-----------------------------------------------------------
                           0.661   slack (MET)

worst slack max 36.16
worst slack min 0.66
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   1.43 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.10 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.33 setup skew

clk period_min = 8.85 fmax = 113.05
=== CHECK TYPES (empty = none) ===
```

6.9 Corner ss_n40C_1v28 (slow)

Setup slack **+10.609 ns** and hold slack **+1.633 ns**: both checks are met, with no rule violations. Minimum period 34.39 ns (fmax 29.1 MHz), so the 45 ns clock has 1.3x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v28
=== SETUP ===
Startpoint: core.CPU_is_slli_a3$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          48.122   data required time
                          37.513   data arrival time
-----------------------------------------------------------
                          10.609   slack (MET)

=== HOLD ===
Startpoint: core.CPU_reset_a1$_DFF_P_
Endpoint:   core.CPU_reset_a2$_DFF_P_
Path Group: clk    Path Type: min
                           6.763   data arrival time
                           5.131   data required time
-----------------------------------------------------------
                           1.633   slack (MET)

worst slack max 10.61
worst slack min 1.63
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   5.64 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -4.37 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   1.27 setup skew

clk period_min = 34.39 fmax = 29.08
=== CHECK TYPES (empty = none) ===
```

6.10 Corner ss_n40C_1v35 (slow)

Setup slack **+21.351 ns** and hold slack **+1.309 ns**: both checks are met, with no rule violations. Minimum period 23.65 ns (fmax 42.3 MHz), so the 45 ns clock has 1.9x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v35
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          47.367   data required time
                          26.016   data arrival time
-----------------------------------------------------------
                          21.351   slack (MET)

=== HOLD ===
Startpoint: core.CPU_reset_a1$_DFF_P_
Endpoint:   core.CPU_reset_a2$_DFF_P_
Path Group: clk    Path Type: min
                           4.913   data arrival time
                           3.605   data required time
-----------------------------------------------------------
                           1.309   slack (MET)

worst slack max 21.35
worst slack min 1.31
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   4.00 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -3.14 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.86 setup skew

clk period_min = 23.65 fmax = 42.29
=== CHECK TYPES (empty = none) ===
```

6.11 Corner ss_n40C_1v40 (slow)

Setup slack **+25.962 ns** and hold slack **+1.113 ns**: both checks are met, with no rule violations. Minimum period 19.04 ns (fmax 52.5 MHz), so the 45 ns clock has 2.4x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v40
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          46.993   data required time
                          21.031   data arrival time
-----------------------------------------------------------
                          25.962   slack (MET)

=== HOLD ===
Startpoint: core.CPU_reset_a1$_DFF_P_
Endpoint:   core.CPU_reset_a2$_DFF_P_
Path Group: clk    Path Type: min
                           4.076   data arrival time
                           2.963   data required time
-----------------------------------------------------------
                           1.113   slack (MET)

worst slack max 25.96
worst slack min 1.11
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   3.29 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -2.58 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.71 setup skew

clk period_min = 19.04 fmax = 52.53
=== CHECK TYPES (empty = none) ===
```

6.12 Corner ss_n40C_1v44 (slow)

Setup slack **+28.689 ns** and hold slack **+1.009 ns**: both checks are met, with no rule violations. Minimum period 16.31 ns (fmax 61.3 MHz), so the 45 ns clock has 2.8x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v44
=== SETUP ===
Startpoint: core.CPU_is_slli_a3$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          46.741   data required time
                          18.052   data arrival time
-----------------------------------------------------------
                          28.689   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           3.741   data arrival time
                           2.733   data required time
-----------------------------------------------------------
                           1.009   slack (MET)

worst slack max 28.69
worst slack min 1.01
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   2.86 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -2.26 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.59 setup skew

clk period_min = 16.31 fmax = 61.31
=== CHECK TYPES (empty = none) ===
```

6.13 Corner ss_n40C_1v60 (slow)

Setup slack **+34.807 ns** and hold slack **+0.681 ns**: both checks are met, with no rule violations. Minimum period 10.19 ns (fmax 98.1 MHz), so the 45 ns clock has 4.4x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v60
=== SETUP ===
Startpoint: core.CPU_is_slli_a3$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          46.241   data required time
                          11.434   data arrival time
-----------------------------------------------------------
                          34.807   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           2.510   data arrival time
                           1.829   data required time
-----------------------------------------------------------
                           0.681   slack (MET)

worst slack max 34.81
worst slack min 0.68
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   1.92 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -1.53 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.39 setup skew

clk period_min = 10.19 fmax = 98.11
=== CHECK TYPES (empty = none) ===
```

6.14 Corner ss_n40C_1v76 (slow)

Setup slack **+37.681 ns** and hold slack **+0.508 ns**: both checks are met, with no rule violations. Minimum period 7.32 ns (fmax 136.6 MHz), so the 45 ns clock has 6.1x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: ss_n40C_1v76
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[3][29]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.986   data required time
                           8.305   data arrival time
-----------------------------------------------------------
                          37.681   slack (MET)

=== HOLD ===
Startpoint: core.CPU_reset_a2$_DFF_P_
Endpoint:   core.CPU_reset_a3$_DFF_P_
Path Group: clk    Path Type: min
                           1.892   data arrival time
                           1.383   data required time
-----------------------------------------------------------
                           0.508   slack (MET)

worst slack max 37.68
worst slack min 0.51
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   1.30 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.01 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.29 setup skew

clk period_min = 7.32 fmax = 136.63
=== CHECK TYPES (empty = none) ===
```

6.15 Corner tt_025C_1v80 (typical)

Setup slack **+40.585 ns** and hold slack **+0.328 ns**: both checks are met, with no rule violations. Minimum period 4.41 ns (fmax 226.5 MHz), so the 45 ns clock has 10.2x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: tt_025C_1v80
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.755   data required time
                           5.170   data arrival time
-----------------------------------------------------------
                          40.585   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.254   data arrival time
                           0.926   data required time
-----------------------------------------------------------
                           0.328   slack (MET)

worst slack max 40.59
worst slack min 0.33
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.87 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.69 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.18 setup skew

clk period_min = 4.41 fmax = 226.50
=== CHECK TYPES (empty = none) ===
```

6.16 Corner tt_100C_1v80 (typical)

Setup slack **+40.673 ns** and hold slack **+0.331 ns**: both checks are met, with no rule violations. Minimum period 4.33 ns (fmax 231.1 MHz), so the 45 ns clock has 10.4x frequency margin at this corner.

```
PARASITICS: global_routing
CORNER: tt_100C_1v80
=== SETUP ===
Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
Endpoint:   core.CPU_Xreg_value_a4[26][17]$_SDFFE_PP0P_
Path Group: clk    Path Type: max
                          45.735   data required time
                           5.062   data arrival time
-----------------------------------------------------------
                          40.673   slack (MET)

=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
Endpoint:   core.CPU_rd_a3[0]$_DFF_P_
Path Group: clk    Path Type: min
                           1.224   data arrival time
                           0.893   data required time
-----------------------------------------------------------
                           0.331   slack (MET)

worst slack max 40.67
worst slack min 0.33
wns max 0.00
tns max 0.00
wns min 0.00
tns min 0.00
Clock clk
   0.85 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.67 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.18 setup skew

clk period_min = 4.33 fmax = 231.13
=== CHECK TYPES (empty = none) ===
```

7. Conclusion

The repaired, routed vsdbabysoc design meets timing at all 16 sky130_fd_sc_hd corners with a 45 ns clock. Setup slack is positive everywhere (minimum +10.609 ns at ss_n40C_1v28), hold slack is positive everywhere (minimum +0.203 ns at ff_n40C_1v95), and no slew/capacitance violations remain.

Points to state clearly when presenting these results

1. **The clock is relaxed to 45 ns (about 22 MHz).** The 11 ns clock cannot be met in the slowest corners (the minimum period is 34.39 ns at ss_n40C_1v28). Setup passes here because of the relaxed clock, and this needs your mentor's acceptance.

2. **The repair is a post-route, in-place estimate.** 162 instances were resized and 50 buffers inserted. The new buffers are placed, but their wires were not re-routed, and the timing numbers use *global_routing* parasitics rather than detailed-route extraction. A full re-run of the flow with a slow-corner library is the proper confirmation.

3. **Original data is preserved.** The fix lives in a new file, 5_route_repaired.odb. The original routed database and the 11 ns reports are unchanged.

4. **Corner coverage.** The 16 corners analysed are the Liberty files available in skywater-pdk-libs-sky130_fd_sc_hd/timing. If your course or mentor requires a specific subset, the table above can be filtered accordingly.